MIL-C-60690A(MU)
3 April 1968
SUPERSEDING
MIL-C-60690(MU)
20 April 1967

MILITARY SPECIFICATION

CHRONOGRAPH AUTOMATIC RELIABILITY RATER: 10552370

1. SCOPE

1.1 This specification covers a testing device utilized with the M36 Radar Chronograph System and designated as the Chronograph Automatic reliability Rater, herein referenced as CARR.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction, M18, Manufacturing Process and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

US Army, Frankford Arsenal

F10552370	Chronograph Automatic Reliability Rater

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PACKAGING DATA SHEET

10552370 — Packaging of Chronograph Automatic Reliability Rater

(Copies of Specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The CARR shall be manufactured in accordance with Drawing F10552370 and drawings pertaining thereto.

3.2 General specifications.- The CARR shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship
e. Soldering electrical connections

3.2.2 MIL-C-46357

a. Wiring
b. Threaded fasteners
c. Marking
d. Finishes
e. Etched circuit assemblies

3.3 Environmental

3.3.1 Orientation.- The CARR shall have its cover completely open as would be normal during operational use, such as to give complete access to the front panel controls, during the performance of 3.3.4, 3.3.5, 3.3.6, 3.3.7, 3.3.8, 3.3.9, and 3.3.11.

3.3.2 Shock.- With the cover closed and securely latched and the cable inside the cover attached to its holding clips, the CARR shall be capable of withstanding three shocks of 15g, 11 millisecond duration in each of the following directions:

a. Bottom down
b. Front down
c. Right side down
d. Back down
e. Left side down
f. Top down

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Subsequent to the above shock, the CARR shall not have been damaged or adversely affected and shall meet the requirements of para. 3.4.

3.3.3 Vibration.- With the cover closed and securely latched and the cable inside the cover attached to its holding clips, the CARR shall be capable of being vibrated in a simple harmonic motion having an amplitude of 0.03 inch (0.06 inch maximum total excursion) with the frequency being varied uniformly between the limits of 10 and 55 cycles per second. The entire frequency range, from 10 to 55 cps and return to 10 cps shall be traversed in 1 minute ± 15 seconds. This motion shall be applied for a period of 30 minutes in each of three (3) mutually perpendicular directions (x, y, z axes) for a total of 1.5 hours. Subsequent to vibration, the CARR shall not have been damaged or adversely affected and shall meet the requirements of para. 3.4.

3.3.4 Rain.- The CARR shall be capable of withstanding 5 hours of continuous exposure to heavy rainfall in the amount of 2.0 inches/hour with a wind speed of 35 knots at a water temperature of plus 85°F. Subsequent to such rainfall, the CARR shall have no evidence of water penetration inside the volume housing the electrical circuitry and shall not have been damaged or adversely affected and shall meet the requirements of para. 3.4.

3.3.5 Humidity.- The CARR shall be capable of withstanding 4 hours of continuous exposure to a relative humidity of 90 to 95 percent at a temperature of plus 40° ± 2°C. During and subsequent to the above humidity, the CARR shall not have been damaged or adversely affected, and shall meet the requirements of paragraph 3.4.

3.3.6 Sand and Dust.- The CARR shall be capable of withstanding 3 hours of continuous exposure to blowing sand of particle size 0.01 to 1.00 mm and blowing dust or particle size 0.001 to 0.01mm with a density between 0.1 to 0.5 gram per cubic foot of sand and a density between 0.1 to 0.5 gram per cubic foot of dust at a temperature of plus 25°C, a relative humidity between 15 to 30%, and a minimum wind velocity of 15 knots. Subsequent to sand and dust exposure, the CARR shall not have been damaged or adversely affected and shall meet the requirements of paragraph 3.4.

3.3.7 Salt Spray.- The CARR shall be capable of withstanding 48 hours of continuous exposure to a fine spray of 5% salt solution concentration with a specific gravity of from 1.0268 to 1.0413, a PH between 6.5 and 7.2, and a temperature between plus 93°F and plus 97°F. Subsequent to salt spray, the CARR shall not have been damaged or adversely affected and shall meet the requirements of paragraph 3.4.

3.3.8 Altitude.

3.3.8.1 Operating.- The CARR shall be capable of operating during 4 hours of continuous exposure to an altitude of 10,000 ft. at a temperature of plus 75°F plus or minus 15 degrees Fahrenheit. During and subsequent to such exposure, the CARR shall not have been damaged or adversely affected and shall meet the requirements of paragraph 3.4.

3.3.8.2 Non-Operating.- The CARR shall be capable of 4hours of continuous exposure to an altitude of 50,000 ft at a temperature of minus 65°F ± 5°F. Subsequent to such exposure, the CARR shall not have been damaged or adversely affected and shall meet the requirements of para.3.4.

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3.3.9 Fungus.- The CARR shall be capable of 28 days exposure to a composite spore suspension consisting of the following cultures: (a) Chaetomium globosum, (b) Aspergillus versicalor, (c) Aspergillus flavus, (d) Pencillium luteum, (e) Rhizopus nigricans. The composite spore suspension shall be incubated at an ambient temperature of between plus 77°F and plus 84°F with an RH of 97 ± 2%. Subsequent to the above fungus exposure, the CARR shall not have been damaged or adversely affected and shall meet the requirements of paragraph 3.4.

3.3.10 Storage temperatures.- The CARR shall be capable of withstanding temperature stabilization at plus 160°F and minus 80°F for a period of 8 hours at each temperature extreme. Subsequent to temperature stabilization at plus 75 ± 15°F after such temperature extreme, the CARR shall not have been damaged or adversely affected and shall meet the requirements of paragraph 3.4.

3.3.11 Operating temperature.- The CARR shall be capable of 8 hours continuous operation at plus 125°F with 360 BTU/sq. foot/hour with a wind velocity of 6 knots. The CARR shall be capable of 8 hours continuous operation at minus 50°F. During and subsequent to the above temperature extremes, plus 125°F with 360 BTU/sq. foot/hour and minus 50°F, the CARR shall not have been damaged or adversely affected and shall meet the requirements of paragraph 3.4 at least once at the beginning, middle, and ending of each 8 hour temperature extreme.

3.4 Performance

3.4.1 Power supply.- All requirements of 3.4 shall be met with the 22V power input varying between the limits of 21.4 to 22.6 volts.

3.4.1.1 The initial settings of the timing circuits with the selected resistors shall be accomplished with the power input at +22.0V ± .1V on J1B.

3.4.2 Mike Trigger output.- With a 1K ± 1% resistor load between J1A and ground, the voltage at J1A shall be between 4.5% to 5.4% of the input voltage at J1B (+22.0V nominal).

3.4.2.1 When the START Switch S1 is held depressed, J1A shall be at ground potential. When the START Switch S1 is released, J1A shall return to the voltage specified in 3.4.2.

3.4.3 Initial delay circuit.- With a 1k ± 1% resistance load between J1A and ground, the time interval from the instant that J1A is at ground potential until the collector of transistor Q1 falls negatively to the collector-to-emitter saturation voltage of Q1, shall be as specified in TABLE I for each of the delay positions of Switch S3.

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TABLE I

Switch S3 Delay Position	Delay Interval	Tolerance At 70°F ± 7°F
1	158.0 Milliseconds	± 1 Millisecond
2	200.8 Milliseconds	± 1 Millisecond
3	243.6 Milliseconds	± 1 Millisecond
4	286.4 Milliseconds	± 1 Millisecond
5	329.2 Milliseconds	± 1 Millisecond
6	372.0 Milliseconds	± 1 Millisecond
7	414.8 Milliseconds	± 1 Millisecond
8	457.6 Milliseconds	± 1 Millisecond
9	500.4 Milliseconds	± 1 Millisecond
10	543.2 Milliseconds	± 1 Millisecond

3.4.4 Delay Gate Circuit.- At all of the 10 numbered positions of Delay Switch S3, the time interval from the instant the collector of Q1 falls negatively to the collector-emitter saturation voltage of Q1 to the instant the collector of Q3 falls negatively to the collector-emitter saturation voltage of Q3, shall be 68 milliseconds with a tolerance of ± 2.5 Millisecond at 70°F ± 7°.

3.4.5 Temperature requirements of initial delay circuit and delay gate circuit.- Over the temperature extremes of minus 50°F to plus 120°F, with 360 BTU/Sq. ft/hr of solar radiation at 120°F only, the tolerance on the requirements of 3.4.3 shall be ± 9 milliseconds and on the requirements of 3.4.4 shall be ± 4 milliseconds.

3.4.6 Oscillator Circuit.

3.4.6.1 Wave form.- With Delay Switch S3 in "CONT" position, and with Switch S4 in the "OSC" position, the output at J2A shall be a square wave having an amplitude of 60 millivolts ± 20 millivolts superimposed on a 90 ± 30 mv d.c. level. The percentage of the "ON" time interval at the 90% point and the "OFF" time interval at the 10% point, as shown in Figure 1, shall be a minimum of 25% of the period.

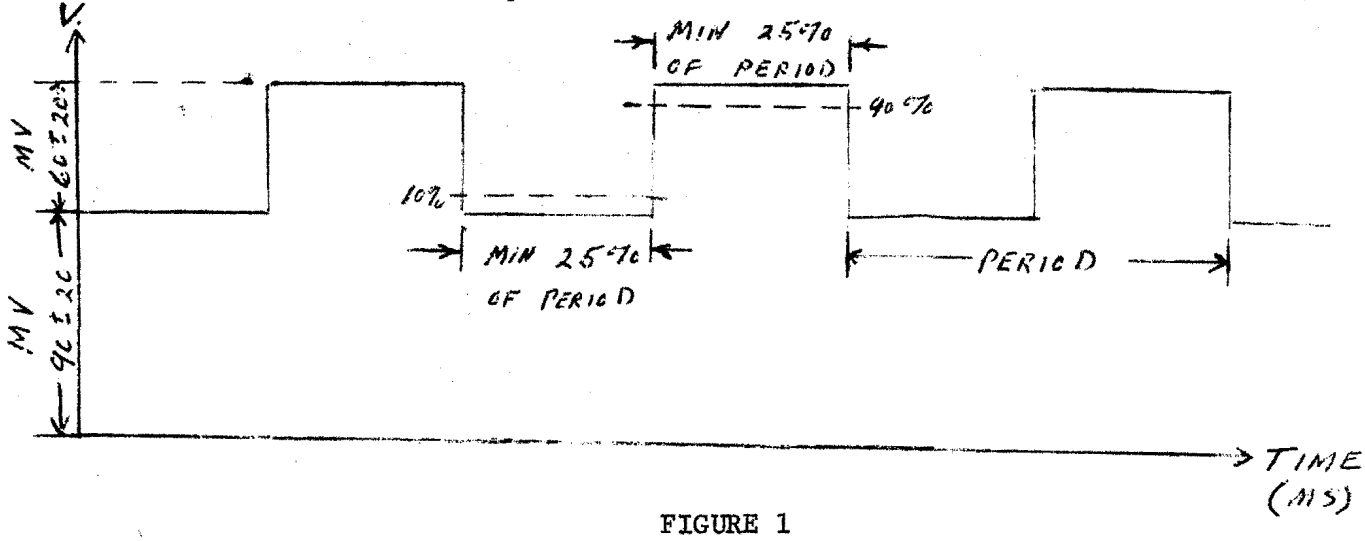


FIGURE 1

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3.4.6.2 Frequency.- The frequency of the output square wave for each position of the Velocity Selector Switch S2, shall be as specified in Table II.

TABLE II

Velocity Position	Frequency
70-325	16.000 KHZ ± .012%
300-400	25.000 KHZ ± .012%
375-525	30.000 KHZ ± .012%
500-700	40.000 KHZ ± .012%
650-800	50.000 KHZ ± .012%
750-1000	65.000 KHZ ± .012%
925-1150	75.000 KHZ ± .012%
1100-1400	95.648 KHZ ± .012%
1400-1590	110.069 KHZ ± .012%
1525-1860	115.000 KHZ ± .012%

3.4.6.3 The conditions of 3.4.6 shall be accomplished within a maximum of 10 seconds when switching from any one VELOCITY position to any other VELOCITY position.

3.4.7 Output conditions.- With a 1K ± 1% resistor connected to J1A, the following conditions shall exist:

3.4.7.1 With Switch S4 in the METER, OSC and OFF positions, there shall be no current flowing thru the METER in any position of the DELAY selector switch and VELOCITY selector switch.

3.4.7.2 The meter shall indicate between 400 and 500 milliamperes when +1.80 ± 0.05 vdc is applied through a 1.5K ± 1% resistor to J2A with S4 in METER position only.

3.4.7.3 With DELAY Switch S3 in DELAY Selector positions 1 thru 10, there shall be no voltage at J2A when S4 is in the OSC, METER or OFF positions.

3.4.7.4 With Switch S4 in the OSC position and the DELAY Switch S3 in each position 1 thru 10, the output waveshape when START Switch S1 is depressed shall be as shown in Figure 2 where T_d corresponds to the time interval given in TABLE I for each respective DELAY position 1 thru 10. The t_o point shall be the time when J1A with 1K ± 1% resistor load goes to ground potential. The frequency of the waveshape inside the envelope shall be as called out in Table II for each position of the VELOCITY Switch.

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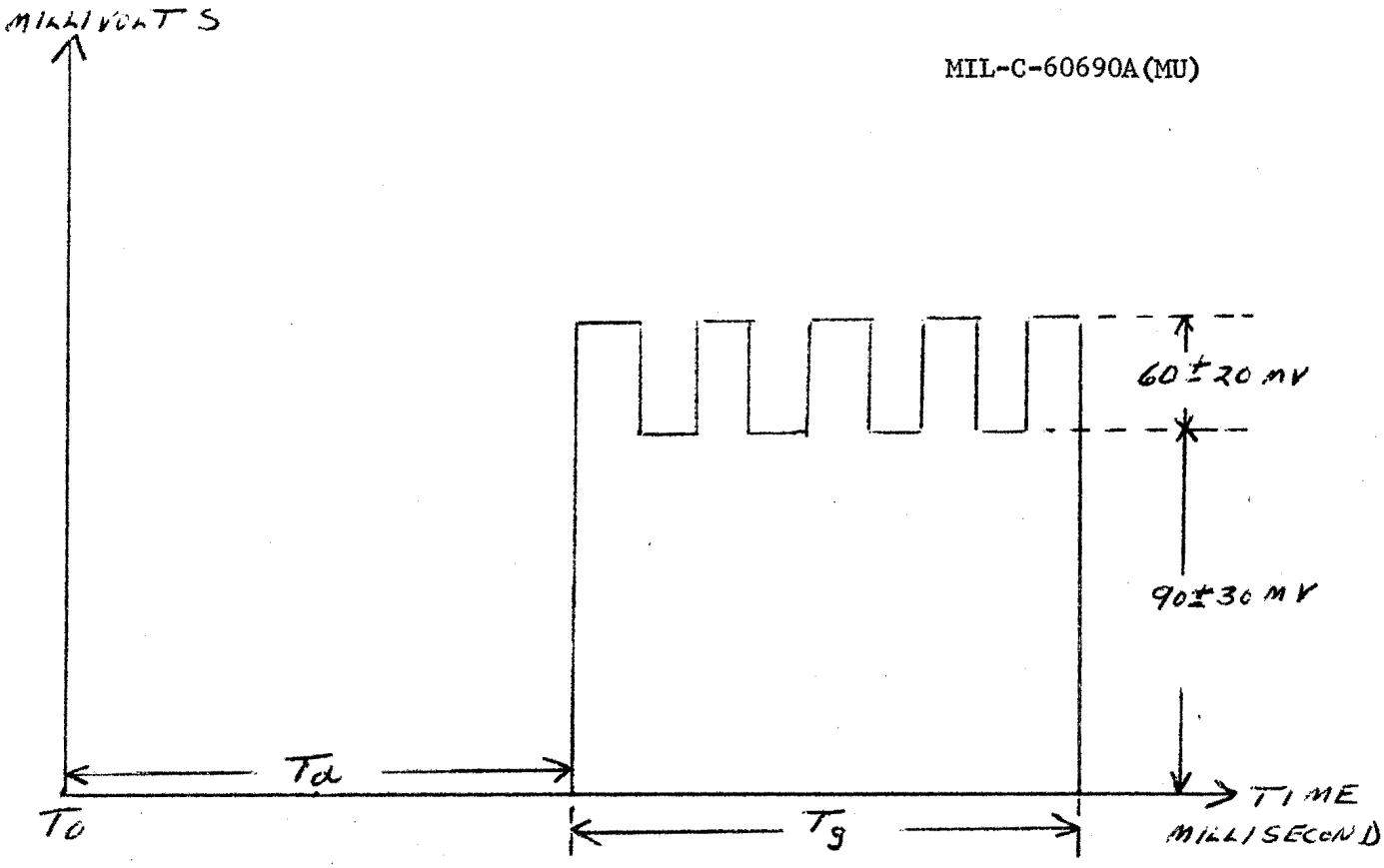


FIGURE 2

3.4.7.5 The tolerance on time t_d and t_g over the environmental temperature extremes shall be as in 3.4.5. The tolerance on the square wave shall be as given in 3.4.6 and 3.4.6.1.

3.5 Reliability Run-In.- In order to maximize component reliability, each CARR shall, prior to acceptance, be subjected to the following run-in conditions:

3.5.1 Temperature run-in.- The CARR, without power applied, will be temperature stabilized at plus 125°F and minus 25°F. At each temperature extreme, after temperature stabilization, power will be applied and each unit will immediately be tested to the performance requirements of 3.4 for a period of three (3) hours with a minimum of one (1) complete test of 3.4 occuring at the beginning, middle, and end of the three (3) hour period. Subsequent to the above tests, the CARR shall not have been damaged or adversely affected and shall meet the requirements of paragraph 3.4.

3.5.2 Vibration run-in.- The CARR with the cover closed and securely latched and the cable inside the cover attached to its holding clips shall be vibrated in a simple harmonic motion having an amplitude of 0.03 inch (0.06 inch maximum total excursion). The frequency shall be uniformly varied between the limits of 10 and 55 cycles per second. The entire frequency range, from 10 to 55 cps and return to 10 cps, shall be traversed in 1 minute ± 15 seconds. This motion shall be applied for a period of 30 minutes in each of three (3) mutually perpendicular directions (x, y, z axes) for a total of 1.5 hours. Subsequent to vibration, the CARR shall not have been damaged or adversely affected and shall meet the requirements of paragraph 3.4.

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3.5.3 Failures during reliability tests.- Replacement of defective components and correction of workmanship defects may be made during run-in under the following conditions:

a. Any part, module, component or subassembly that is used as a replacement shall have previously passed environmental tests consistent with the reliability run-in environmental requirements of the CARR.

b. Any replacement part shall have been exposed for at least one (1) hour at the extreme temperature being run immediately previous to being used as a replacement.

c. After a correction has been accomplished, at least one (1) hour of operation with a minimum of one (1) test to the requirements of 3.4 shall be successfully accomplished.

d. A failure during para 3.5.2 shall necessitate a complete re-running of the conditions of para 3.5.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) inspection.- When required by the contract, an initial production sample shall be submitted to a Government approved facility in accordance with the contract schedule for test and evaluation. The sample shall consist of 3 units. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures, as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to all requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials. After inspection by the contractor and provisional acceptance at source by the Government, the sample shall be inspected at a Government facility or such other facility as specified in the contract, for all the requirements of the drawings and specifications.

4.3 Inspection provisions.

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4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and test.

4.3.2.1 Component and subassemblies.- All components and subassemblies to be inspected in accordance with the inspection provisions contained in the applicable Quality Assurance Provisions of MIL-F-13926.

4.3.3 Acceptance and rejection: Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.- At the beginning of production, three units shall be selected by a Government representative and then two from each 50 produced. The sample shall have successfully met all other examinations and acceptance tests specified herein, and shall meet the following requirements and tests:

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
401.	Fabrication	3.1	Applicable dwgs & Specs - Visual
402.	General specifications	3.2	MIL-F-13926 & MIL-F-46357
403.	Shock	3.3.2	4.6.1.1
404.	Vibration	3.3.3	4.6.1.2
405.	Rain	3.3.4	4.6.1.3
406.	Humidity	3.3.5	4.6.1.4
407.	Sand and Dust	3.3.6	4.6.1.5
408.	Salt spray	3.3.7	4.6.1.6
409.	Altitude-operating	3.3.8.1	4.6.1.7
410.	Altitude-non-operating	3.3.8.2	4.6.1.8
411.	Storage temperatures	3.3.10	4.6.1.9.1
412.	Operating temperatures	3.3.11	4.6.1.9.2

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor, and resubmitted samples have been approved (see 4.3.2).

4.5 Inspection equipment.- Except as other wise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

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4.5.1 Government furnished inspection equipment.- Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Contractor design.- The contractor shall design and supply inspection equipment compatible with "Test Methods and Procedures" specified in Paragraph 4.6 of this specification, and with the "Component Inspection" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of the test equipment.

4.6 Test methods and procedures.

4.6.1 Environmental.- Test equipment and test facilities utilized in the following environmental tests shall be in accordance with "Test Facilities requirements of MIL-F-13926 and conditions specified in 3.3.1.

4.6.1.1 Shock.- The shock test shall be performed as outlined in MIL-STD-202, Method 205, test condition A. Subsequent to the 6 shocks, the CARR shall be visually examined for physical damage, then tested as specified in 4.6.2.

4.6.1.2 Vibration.- The vibration test shall be performed as outlined in MIL-STD-202, Method 201. Subsequent to the 1.5 hour vibration, the CARR shall be visually examined for physical damage, then tested as specified in 4.6.2.

4.6.1.3 Rain.- The rain test shall be performed as outlined in MIL-F-13926, Procedure I with the following exceptions: Five (5) hours of continuous rainfall at a rate of 2 inches/hour and wind velocity of 35 knots. Temperature of the water shall be maintained at approximately 85°F. Subsequent to the rain test, the CARR shall be examined for any water penetration and physical damage, then tested as specifiedin 4.6.2.

4.6.1.4 Humidity.- The humidity test shall be performed as outlined in MIL-STD-202, Method 103 for four (4) hours. Subsequent to this test, the CARR shall be examined for damage then tested as specified in 4.6.2.

4.6.1.5 Sand and dust.- The sand and dust test shall be performed as outlined in MIL-STD-202, Method 110, with the following conditions: The sand and dust particle sizes, density, wind velocity, temperature, time duration and relative humidity shall be as specified in 3.3.6. Subsequent to this test, the CARR shall be visually examined for damage then tested as specified in 4.6.2.

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4.6.1.6 Salt spray.- The salt spray test shall be performed as outlined in MIL-STD-202, Method 101, Condition B. Subsequent to this test, the CARR shall be visually examined for damage, then tested as specified in 4.6.2.

4.6.1.7 Altitude-operating.- The altitude test shall be performed as outlined in MIL-F-13926, Procedure I at an altitude of 10,000 feet and as specified in 3.3.8.1. The CARR shall be exposed to the above condition for a time period required to allow all entrapped air in the chamber to escape as evidenced by chamber pressure stabilization. The barometric pressure shall then be maintained to simulate a 10,000 foot (20.58 inches of mercury) altitude condition. While the CARR is maintained at this pressure for four (4) hours, the CARR shall be subjected to the tests specified in 4.6.2.

4.6.1.7.1 Altitude-non-operating.- The CARR shall be subjected to barometric pressure outlined in MIL-F-13926, Procedure II for four (4) hours as specified in 3.3.8.2. Subsequent to this test, the CARR shall be visually examined for damage, then tested as specified in 4.6.2.

4.6.1.8 Fungus.- One (1) CARR shall be subjected to the fungus test outlined in MIL-F-13926. At the end of the 28 day incubation period, the CARR shall be tested as specified in 4.6.2 for compliance with 3.3.9. Subsequent acceptance of the CARR shall be by certification.

4.6.1.9 Temperature cycling.

4.6.1.9.1 Storage temperatures.- This test shall be performed as outlined in MIL-F-13926, Procedure II, and thermally stabilized at +160°F for eight (8) hours and at -80°F for eight (8) hours. The CARR shall be returned to room ambient temperature +75 $\pm$ 15°F, visually examined for damage then tested as specified in 4.6.2.

4.6.1.9.2 Operating temperatures.- This test shall be performed as outlined in MIL-F-13926, Procedure I, Condition F with the following conditions. Thermally stabilized at +125°F with 360 BTU/sq. ft/hr introduced, and with wind velocity of 6 knots maintained for eight (8) hours. The CARR shall be tested as specified in 4.6.2 during the first hour, at the four (4) hour period and within the last hour of the eight (8) hour period. The temperature shall then be reduced to -50°F. When thermally stabilized at this temperature, the CARR shall be tested as specified in 4.6.2 at the three (3) time periods specified above.

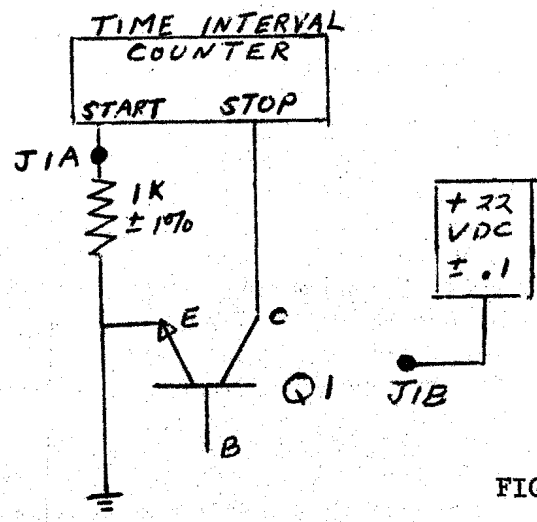
4.6.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be performed on a100% basis. The power inputs shall be as specified in 3.4.1 and 3.4.1.1.

4.6.2.1 Mike trigger output.- Apply +22.0 $\pm$ 0.1 vdc to J1B. Connect a 1K ohm $\pm$ 1% resistor from J1-A to ground. Measure the voltage drop across the resistor. The voltage reading shall be between 4.5% and 5.4% of the +22 voltage input.

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4.6.2.1.1 Depress START switch S1. The measured voltage across the resistor shall be at ground potential. Release S1, the voltage measured shall return to the value specified in 4.6.21 Repeat the above tests with power inputs at +21.4 ± 0.1 volts and at +22.6 ± 0.1 volts. The measured outputs shall be between 4.5% and 5.4% of the power inputs specified above.

4.6.2.2 Initial delay circuit.- Connect the power supply and a time interval counter as shown in Figure 3. Depress S1. The time interval delay at each switch S3 position shall be as shown in Figure 3 and Table I. The tolerance on each delay interval shall be ± 1 millisecond



Delay Interval		S3 Position	
158.0	MS	1	0
200.8	"	2	0
243.6	"	3	0
286.4	"	4	0
329.2	"	5	0
372.0	"	6	0
414.8	"	7	0
457.6	"	8	0
500.4	"	9	0
543.2	"	10	0
Continuous		11	0

DELAY SWITCH S3

FIGURE 3

Repeat the test in Figure 3 with inputs of +21.4 ± 0.1 volts and +22.6 ± 0.1 volts. The outputs shall be within the ± 1 millisecond tolerance.

4.6.2.3 Delay gate circuit.- Make all connections as shown in Figure 4. The delay gate time interval shall read 68 ± 2.5 millisecond when S1 is depressed for all ten (10) positions of Switch S3. Repeat above test using +21.4 ± 0.1 volts and +22.6 ± 0.1 volts in place of +22 ± 0.1 volts.

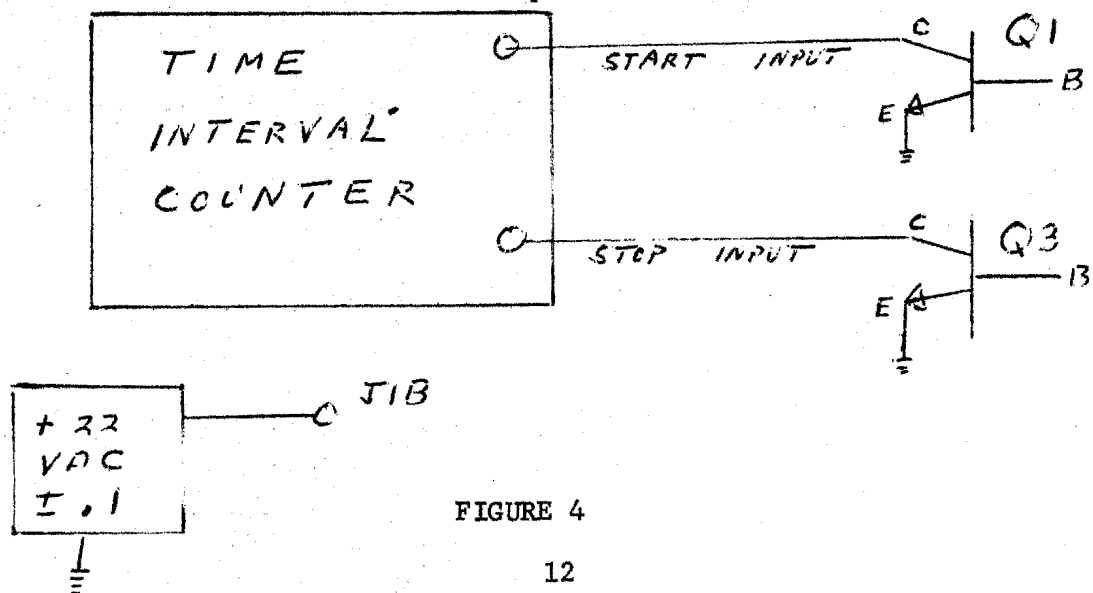


FIGURE 4

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4.6.2.4 On the special samples specified in 4.4 repeat the tests specified in 4.6.2.2 and 4.6.2.3 at the temperatures specified in 3.4.5. When thermally stabilized at minus 50°F, and at plus 120°F with 360 BTU/sq. ft/hr of solar radiation introduced, the results obtained shall be as specified in 4.6.2.2 and 4.6.2.3 with the following exceptions. The tolerance on the initial delay interval shall be ± 9 milliseconds and on the delay gate interval shall be ± 4 milliseconds.

4.6.2.5 Oscillator circuit.- Connect the power supply, oscilloscope and frequency counter as shown in Figure 5A. Set Delay Switch S3 in "CONT" position and S4 in "OSC" position. The square wave output at J2 shall be 60 ± 20 millivolts. The percentage at the 90% point of "ON" time and the 10% point of "OFF" time shall be 25% as specified in 3.4.6.1 and Figure 1. Operate the velocity switch to the positions shown in Figure 5B. The measured frequency shall be as indicated with respect to each position. Switch from the 1525-1800 velocity position to all of the other nine (9) velocity positions. The oscillator shall be stable at the new position within ten (10) seconds and shall meet the requirements of 3.4.6.1 in all ten (10) positions.

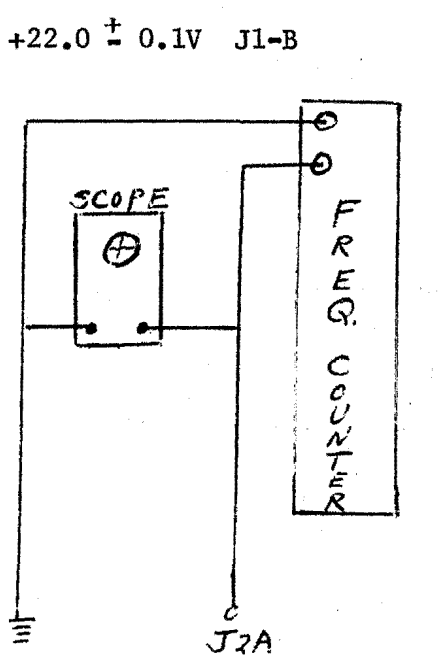


FIGURE 5A

SCOPE	VELOCITY	S2
16Khz ± .012%	70 - 325	0
25K" "	300 - 400	0
30K" "	375 - 525	0
40K" "	500 - 700	0
50K" "	650 - 800	0
65K" "	750 - 1000	0
75K" "	925 - 1150	0
95.648Khz "	1100 - 1400	0
110.069 " "	1400 - 1590	0
115 " "	1525 - 1860	0

FIGURE 5B

Repeat the above tests with +22.6 ± 0.1 volts and +21.4 ± 0.1 volts. The outputs shall meet the specified requirements of 3.4.6.1.

4.6.2.6 Output conditions.- Connect the +22.0 ± 0.1 volt power supply to J1-B and the 1K ohm ± 1% resistor from J1-A to ground. Rotate the delay switch and velocity switch to all positions. The front panel meter shall read zero for each position with S4 in "Meter", "OFF and "OSC" positions.

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a. Set S4 to "METER" position. Apply 1.80 ± 0.05 vdc through a 1.5K ± 1% resistor to J2A. The front panel meter shall indicate the current value specified in 3.4.7.2. Disconnect the power source and resistor.

b. With Switch S4 in "OSC", "METER" and "OFF" positions, rotate the delay switch from 1 through 10. A scope connected from J2-A to ground shall indicate less the 5 millivolts peak-to-peak for each S4 position and each delay switch position.

c. With Switch S4 in "OSC" position and Delay Switch S3 in position 1, depress Start Switch S1. The output wave shape displayed on the oscilloscope (see Figure 6) shall be as specified in Figure 2. Turn Test Switch S5 to position 2. The t_d measured on the time interval counter shall be as shown in Figure 2 and Table I.

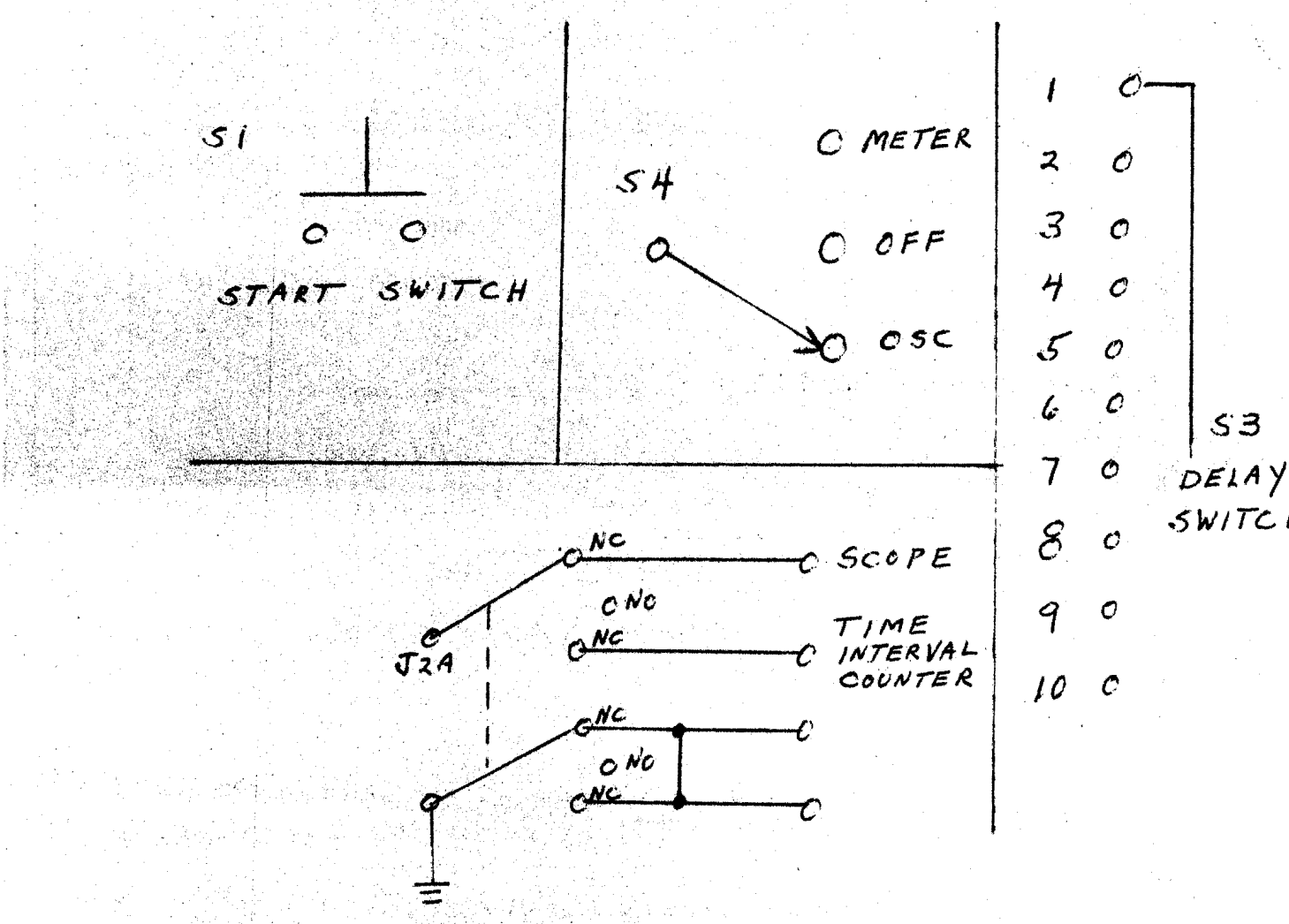


FIGURE 6

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d. On the special samples specified in 4.4 repeat the above tests at the extreme temperatures as outlined in 4.6.2.4. The tolerance on t_d shall be $\pm$ 9 milliseconds and on t_g shall be $\pm$ 4 milliseconds.

4.6.3 Reliability run-in.- The following run-in shall be performed on a 100% basis.

4.6.3.1 Temperature run-in.- The CARR shall be placed in a chamber and stabilized at plus 125°F. Power shall be applied and maintained for three (3) hours. The CARR shall be subjected to all of the tests outlined in 4.6.2 at the beginning of the three (3) hour period, in the middle and near the end of the period. Subsequent to the test, the CARR shall be visually examined for damage. Repeat the above test at minus 25°F. The CARR shall meet the performance requirements of 4.6.2 at both temperature extremes.

4.6.3.2 Vibration run-in.- The CARR shall be subjected to the vibration test as specified in 4.6.1.2. Subsequent to the test, the CARR shall be visually examined for damage and shall meet the performance requirements of 4.6.2.

4.6.3.3 Failures.- Failures occurring during run-in shall not be cause for rejection but shall be corrected as outlined in 3.5.3. A minimum of 1.5 hour run-in at the environment in which failure occurred shall be performed following each replacement during para 4.6.3.1 and the complete test shall be reperformed for any failure during para 4.6.3.2.

4.6.4 Packaging, packing, and marking.- Quality assurance provisions for packaging, packing, and marking shall be as specified in MIL-P-14232 and Section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet 10552370. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The chronograph automatic reliability rater is used in operational testing of the Radar Chronograph: M36.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of preservation, packaging, and packing.
c. Applicable stock number.
d. Applicable packaging data sheet (see 5.1).

Custodian:

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Preparing activity:

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